A NOTE ON COMPLEX HADAMARD MATRICES AND EXAMPLES OF STEINER ETFS

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Abstract. I will be going over the basics of Complex Hadamard Matrices and do some analysis involving particular Steiner systems.

1. Complex Hadamard Matrices

First let us examine the definition of Hadamard matrices.

Definition 1.1 (Hadamard Matrix)**.** An $n \times n$ matrix $\mathbf{H} \in \mathbb{H}$ with unimodular entries that satisfies

$$\mathbf{H}\mathbf{H}^* = n\mathbf{I}, \tag{1.1}$$

where $\mathbf{I}$ is an $n \times n$ identity matrix, the matrix $\mathbf{H}$ is a Hadamard matrix.

Notice that the implication is the that rows/columns are orthogonal. If $\mathbb{H} = \mathbb{R}$ then $\mathbf{H}$ has entries that are ± 1. For $\mathbb{H} = \mathbb{C}$ then we can utilize the Fourier matrix to construct $\mathbf{H}$.

Definition 1.2 (Discrete Fourier Transform)**.** An n-point Discrete Fourier Transformation (DFT) matrix $\mathbf{W} \in \mathbb{C}^{n\times n}$ is defined as

$$\mathbf{W} = \frac{1}{\sqrt{n}}\left[\omega^{(j-1)(k-1)}\right]_{j,k=1}^{n}, \tag{1.2}$$

where $\omega = e^{-2\pi i/n}$.

Notice that $\mathbf{W}\mathbf{W}^* = \mathbf{I}$; therefore, $\mathbf{W}$ is unitary. If we omit $1/\sqrt{n}$ we may calculate our Hadamard matrix in the following way:

$$\mathbf{H} = \left[\omega^{(j-1)(k-1)}\right]_{j,k=1}^{n}, \tag{1.3}$$

and $\omega = e^{-2\pi i/n}$. Notice that since ω is a primitive nth root of unity, the entries of $\mathbf{H}$ are unimodular; thus $\mathbf{H}$ follows (1.1).

2. Full Columns of Complex Hadamard Matrices

We shall examine the columns of the matrix $\mathbf{H}$ as defined by (1.3) and let

$$\mathbf{H} = \Big[\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_n\Big]. \tag{2.1}$$

Date: October 27, 2023.

Since the column elements are unimodular and orthogonal it can be easily shown that the inner products are the following:

$$\langle \mathbf{h}_k, \mathbf{h}_l \rangle = \begin{cases} n & k = l, \\ 0 & k \neq l. \end{cases} \tag{2.2}$$

This also is a result of $\mathbf{H}$ being a Hadamard matrix. Suppose we wish to find the subset norm $\|\mathbf{h}_k \otimes \mathbf{h}_k\|$, this corresponds to a one element subset. Since it is only for one vector, we may use our standard approach and solve the eigenvalue problem

$$\left(\mathbf{h}_k \mathbf{h}_k^*\right) v = \lambda v.$$

If we let $v = a\mathbf{h}_k$, then we have

$$\begin{aligned} \left(\mathbf{h}_k \mathbf{h}_k^*\right) v &= \lambda v \\ \left(\mathbf{h}_k \mathbf{h}_k^*\right) (a\mathbf{h}_k) &= \lambda \left(a\mathbf{h}_k\right) \\ a\mathbf{h}_k \mathbf{h}_k^* \mathbf{h}_k &= \lambda a \mathbf{h}_k \\ n a \mathbf{h}_k &= \lambda a \mathbf{h}_k \\ na &= \lambda a. \end{aligned} \tag{2.3}$$

Notice that $a = 1$ will yield $v = \mathbf{h}_k$ and $\lambda = n$. On the other if $\lambda = 0$ then $v = \mathbf{0}$ and $a = 0$. Thus the two spectral values are $\{n, 0\}$, next we shall find the multiplicity of each spectral value. We have yet to calculate the matrix $\mathbf{h}_k \otimes \mathbf{h}_k$, but this is not necessary since we may find the trace easily. Recall that $\mathbf{h}_k^T = \left\{\omega^{(j-1)(k-1)}\right\}_{j=1}^{n}$; therefore, the trace is given by

$$\operatorname{Tr}(\mathbf{h}_k \otimes \mathbf{h}_k) = \sum_{j=1}^{n} \omega^{(j-1)(k-1)} \overline{\omega}^{(j-1)(k-1)} = n. \tag{2.4}$$

The trace is also equal to the sum of the eigenvalues, and a $n \times n$ Hermitian matrix will have at most n distinct eigenvalues. Therefore,

$$\operatorname{Tr}(\mathbf{h}_k \otimes \mathbf{h}_k) = \lambda_1 + \lambda_2 + \cdots + \lambda_n = n,$$

since one of the eigenvalues is n, then the remaining $n-1$ eigenvalues are 0. Hence, the subset norm of one column from $\mathbf{H}$ is given by

$$\|\mathbf{h}_k \otimes \mathbf{h}_k\| = \lambda_{max} = n. \tag{2.5}$$

This was a trivial example. Suppose now we find a subset norm for two vectors, $\|\mathbf{h}_1 \otimes \mathbf{h}_1 + \mathbf{h}_2 \otimes \mathbf{h}_2\|$; thus, we will solve the eigenvalue problem

$$\left(\mathbf{h}_1 \otimes \mathbf{h}_1 + \mathbf{h}_2 \otimes \mathbf{h}_2\right) v = \lambda v.$$

Let $v = c_1\mathbf{h}_1 + c_2\mathbf{h}_2$, we have

$$\begin{aligned} \lambda(c_1\mathbf{h}_1 + c_2\mathbf{h}_2) &= \left(\mathbf{h}_1\mathbf{h}_1^* + \mathbf{h}_2\mathbf{h}_2^*\right)(c_1\mathbf{h}_1 + c_2\mathbf{h}_2) \\ &= c_1\left(\mathbf{h}_1\mathbf{h}_1^*\mathbf{h}_1 + \mathbf{h}_2\mathbf{h}_2^*\mathbf{h}_1\right) + c_2\left(\mathbf{h}_1\mathbf{h}_1^*\mathbf{h}_2 + \mathbf{h}_2\mathbf{h}_2^*\mathbf{h}_2\right) \\ \lambda c_1\mathbf{h}_1 + \lambda c_2\mathbf{h}_2 &= n c_1\mathbf{h}_1 + n c_2\mathbf{h}_2 \end{aligned} \tag{2.6}$$

If we equate parts of (2.6) we get a similar result as before $\lambda = \{n, 0\}$. It still needs to be determined the multiplicity of the eigenvalues.
The trace $\operatorname{Tr}(\mathbf{h}_a \otimes \mathbf{h}_a + \mathbf{h}_b \otimes \mathbf{h}_b) = \operatorname{Tr}(\mathbf{h}_a \otimes \mathbf{h}_a) + \operatorname{Tr}(\mathbf{h}_b \otimes \mathbf{h}_b) =$

$$\left(\sum_{j=1}^{n} \omega^{(j-1)(a-1)} \overline{\omega}^{(j-1)(a-1)}\right) + \left(\sum_{j=1}^{n} \omega^{(j-1)(b-1)} \overline{\omega}^{(j-1)(b-1)}\right) = 2n.$$

We can use a similar argument as before, the sum of the eigenvalues equals the trace; hence, there are exactly two eigenvalues equal to n and the remaining $n-2$ eigenvalues are 0. The subset norm for a two element subset is n. We can extend the eigenvalue problem to any sized subset of columns from $\mathbf{H}$ to get

$$\left(\sum_{k\in\mathcal{S}} \mathbf{h}_k \otimes \mathbf{h}_k\right) v = \lambda v.$$

Substitute

$$v = \sum_{k\in\mathcal{S}} c_k \mathbf{h}_k$$

into our eigenvalue problem and couple this with (2.2) we get the following:

$$\sum_{k\in\mathcal{S}} n c_k \mathbf{h}_k = \lambda \sum_{k\in\mathcal{S}} c_k \mathbf{h}_k.$$

As was shown before we get that the only two eigenvalues are n and 0. Next let $|\mathcal{S}| = q$ be the cardinality of the subset of columns. Then the trace becomes

$$\mathrm{Tr}\left(\sum_{k\in\mathcal{S}} \mathbf{h}_k \otimes \mathbf{h}_k\right) = qn = \sum_{k=1}^{q} \lambda_k.$$

Since, the only two eigenvalues are n and 0, there must be q eigenvalues that are n and $n-q$ eigenvalues that are 0. Therefore the norm of a subset of columns of a complex $n \times n$ Hadamard matrix $\mathbf{H}$ constructed using a DFT matrix is always n

3. Clipped Columns of Complex Hadamard Matrices

Suppose we define the index set $\mathcal{J} = \{j \in [n] | j \neq p\}$, where $[n] = \{1, \ldots, n\}$ and p is the row we wish to *clip* from $\mathbf{H}$. Notice that we are only clipping one row; thus, let

$$\tilde{\mathbf{H}} = \left[\mathbf{H}_{j,k}\right]_{j\in\mathcal{J}, k\in[n]} = \left[\tilde{\mathbf{h}}_1, \tilde{\mathbf{h}}_2, \cdots, \tilde{\mathbf{h}}_n\right]. \tag{3.1}$$

As a result $\tilde{\mathbf{H}} \in \mathbb{C}^{(n-1)\times n}$ and each column $\tilde{\mathbf{h}}_k \in \mathbb{C}^{n-1}$. Let us find the inner products,

$$\left\langle \tilde{\mathbf{h}}_k, \tilde{\mathbf{h}}_k \right\rangle = \sum_{j\in\mathcal{J}} \omega^{(j-1)(k-1)} \overline{\omega}^{(j-1)(k-1)} = n-1.$$

Suppose we wish to find the inner product between two different columns. We the following

$$\begin{aligned}
\left\langle \tilde{\mathbf{h}}_a, \tilde{\mathbf{h}}_b \right\rangle &= \left\langle \mathbf{h}_a, \mathbf{h}_b \right\rangle - \omega^{(p-1)(a-1)} \overline{\omega}^{(p-1)(b-1)} \\
&= 0 - \omega^{(p-1)(a-1)} \omega^{-(p-1)(b-1)} \\
&= -\omega^{(a-b)(p-1)}.
\end{aligned}$$

Therefore, the row that is clipped will affect the value of the inner product between two different columns. We have the following inner products

$$\langle \tilde{\mathbf{h}}_k, \tilde{\mathbf{h}}_l \rangle = \begin{cases} n-1 & k = l, \\ -\omega^{(k-l)(p-1)} & k \neq l. \end{cases} \tag{3.2}$$

Notice we get a trivial case when $p = 1$ which corresponds to clipping the first row of $\mathbf{H}$. We get

$$\langle \tilde{\mathbf{h}}_k, \tilde{\mathbf{h}}_l \rangle = \begin{cases} n-1 & k = l, \\ -1 & k \neq l. \end{cases}$$

Suppose we wish to find the subset norm corresponding to one vector, which is $\|\tilde{\mathbf{h}}_k \otimes \tilde{\mathbf{h}}_k\|$. Thus we need to solve the eigenvalue problem

$$(\tilde{\mathbf{h}}_k \tilde{\mathbf{h}}_k^*)v = \lambda v.$$

Make the same type of substitution as we did for non-clipped columns to get

$$(\tilde{\mathbf{h}}_k \tilde{\mathbf{h}}_k^*)(a\tilde{\mathbf{h}}_k) = \lambda(a\tilde{\mathbf{h}}_k)$$
$$a(n-1)\tilde{\mathbf{h}}_k = a\lambda\tilde{\mathbf{h}}_k.$$

This yields the same type of solution $\{n-1, 0\}$, notice that the outer product between any two columns will always yield a matrix in $\mathbb{C}^{(n-1)\times(n-1)}$. This means that there are at most $n-1$ distinct spectral values of the matrix $\tilde{\mathbf{h}}_k \otimes \tilde{\mathbf{h}}_k$. It can be shown that $\mathrm{Tr}(\tilde{\mathbf{h}}_k \otimes \tilde{\mathbf{h}}_k) = n-1$; therefore,

- eigenvalue $\lambda = n-1$ has multiplicity $m_1 = 1$
- eigenvalue $\lambda = 0$ has multiplicity $m_2 = n-2$.

Suppose we find the subset norm for a two element subset, we need to calculate

$$\|\tilde{\mathbf{h}}_a \otimes \tilde{\mathbf{h}}_a + \tilde{\mathbf{h}}_b \otimes \tilde{\mathbf{h}}_b\|.$$

This requires solving the eigenvalue problem

$$\left(\tilde{\mathbf{h}}_a \tilde{\mathbf{h}}_a^* + \tilde{\mathbf{h}}_b \tilde{\mathbf{h}}_b^*\right) v = \lambda v.$$

For this case we have a little more work because the inner products are <u>not</u> 0. If we make a similar substitution as we did in the previous example we get

$$\left(\tilde{\mathbf{h}}_a \tilde{\mathbf{h}}_a^* + \tilde{\mathbf{h}}_b \tilde{\mathbf{h}}_b^*\right)\left(c_1\tilde{\mathbf{h}}_a + c_2\tilde{\mathbf{h}}_b\right) = \lambda\left(c_1\tilde{\mathbf{h}}_a + c_2\tilde{\mathbf{h}}_b\right)$$
$$c_1\tilde{\mathbf{h}}_a\tilde{\mathbf{h}}_a^*\tilde{\mathbf{h}}_a + c_1\tilde{\mathbf{h}}_b\tilde{\mathbf{h}}_b^*\tilde{\mathbf{h}}_a + c_2\tilde{\mathbf{h}}_a\tilde{\mathbf{h}}_a^*\tilde{\mathbf{h}}_b + c_2\tilde{\mathbf{h}}_b\tilde{\mathbf{h}}_b^*\tilde{\mathbf{h}}_b =$$
$$c_1(n-1)\tilde{\mathbf{h}}_a - c_1\omega^{(a-b)(p-1)}\tilde{\mathbf{h}}_b - c_2\omega^{(b-a)(p-1)}\tilde{\mathbf{h}}_a + c_2(n-1)\tilde{\mathbf{h}}_b =$$
$$\left(c_1(n-1) - c_2\omega^{(b-a)(p-1)}\right)\tilde{\mathbf{h}}_a + \left(c_2(n-1) - c_1\omega^{(a-b)(p-1)}\right)\tilde{\mathbf{h}}_b = \lambda\left(c_1\tilde{\mathbf{h}}_a + c_2\tilde{\mathbf{h}}_b\right).$$

We equate corresponding parts of the equation to get

$$\lambda c_1 = c_1(n-1) - c_2\omega^{(b-a)(p-1)}$$
$$\lambda c_2 = c_2(n-1) - c_1\omega^{(a-b)(p-1)}.$$

Divide and cross multiply to get

$$c_1c_2(n-1) - c_1^2\omega^{(a-b)(p-1)} = c_2c_1(n-1) - c_2^2\omega^{(b-a)(p-1)}$$
$$c_1^2\omega^{(a-b)(p-1)} = c_2^2\omega^{(b-a)(p-1)}$$
$$c_1^2 = c_2^2\omega^{2(b-a)(p-1)}$$
$$c_1 = \pm c_2\omega^{(b-a)(p-1)}.$$

If we take our second eigenvalue equation and substitute, we get

$$\lambda c_2 = c_2(n-1) \pm c_2\omega^0.$$

Which becomes,

$$\lambda = n - 1 \pm 1.$$

In this case our spectral values are $\{n, n-2, 0\}$. We still need to determine the multiplicities of these spectral values. Notice that if we find the trace of the sum of outer products we get

$$\begin{aligned}\mathrm{Tr}(\tilde{\mathbf{h}}_a \otimes \tilde{\mathbf{h}}_a + \tilde{\mathbf{h}}_b \otimes \tilde{\mathbf{h}}_b) &= \mathrm{Tr}(\tilde{\mathbf{h}}_a \otimes \tilde{\mathbf{h}}_a) + \mathrm{Tr}(\tilde{\mathbf{h}}_b \otimes \tilde{\mathbf{h}}_b) \\ &= (n-1) + (n-1) \\ &= 2n - 2.\end{aligned}$$

Since the trace is also equal to the sum of the eigenvalues we have

$$\begin{aligned}2n - 2 &= m_1 n + m_2(n-2) \\ &= (m_1 + m_2)n - 2m_2,\end{aligned}$$

where m_1, m_2 are the multiplicities of $n, n-2$ respectively. It can be easily shown...

- eigenvalue $\lambda = n$ has multiplicity $m_1 = 1$,
- eigenvalue $\lambda = n-2$ has multiplicity $m_2 = 1$,
- eigenvalue $\lambda = 0$ has multiplicity $m_3 = n - 3$.

Let us examine a three element subset norm

$$\|\tilde{\mathbf{h}}_a \otimes \tilde{\mathbf{h}}_a + \tilde{\mathbf{h}}_b \otimes \tilde{\mathbf{h}}_b + \tilde{\mathbf{h}}_c \otimes \tilde{\mathbf{h}}_c\|.$$

The corresponding eigenvalue problem is

$$\left(\tilde{\mathbf{h}}_a\tilde{\mathbf{h}}_a^* + \tilde{\mathbf{h}}_b\tilde{\mathbf{h}}_b^* + \tilde{\mathbf{h}}_c\tilde{\mathbf{h}}_c^*\right) v = \lambda v.$$

Make the substitution $v = c_1\tilde{\mathbf{h}}_a + c_2\tilde{\mathbf{h}}_b + c_3\tilde{\mathbf{h}}_c$ and the l.h.s. becomes

$$\begin{aligned}&\left(\tilde{\mathbf{h}}_a\tilde{\mathbf{h}}_a^* + \tilde{\mathbf{h}}_b\tilde{\mathbf{h}}_b^* + \tilde{\mathbf{h}}_c\tilde{\mathbf{h}}_c^*\right)\left(c_1\tilde{\mathbf{h}}_a + c_2\tilde{\mathbf{h}}_b + c_3\tilde{\mathbf{h}}_c\right) = \\ &\qquad c_1\tilde{\mathbf{h}}_a\tilde{\mathbf{h}}_a^*\tilde{\mathbf{h}}_a + c_2\tilde{\mathbf{h}}_a\tilde{\mathbf{h}}_a^*\tilde{\mathbf{h}}_b + c_3\tilde{\mathbf{h}}_a\tilde{\mathbf{h}}_a^*\tilde{\mathbf{h}}_c + \\ &\qquad c_1\tilde{\mathbf{h}}_b\tilde{\mathbf{h}}_b^*\tilde{\mathbf{h}}_a + c_2\tilde{\mathbf{h}}_b\tilde{\mathbf{h}}_b^*\tilde{\mathbf{h}}_b + c_3\tilde{\mathbf{h}}_b\tilde{\mathbf{h}}_b^*\tilde{\mathbf{h}}_c + \\ &\qquad c_1\tilde{\mathbf{h}}_c\tilde{\mathbf{h}}_c^*\tilde{\mathbf{h}}_a + c_2\tilde{\mathbf{h}}_c\tilde{\mathbf{h}}_c^*\tilde{\mathbf{h}}_b + c_3\tilde{\mathbf{h}}_c\tilde{\mathbf{h}}_c^*\tilde{\mathbf{h}}_c = \\ &\tilde{\mathbf{h}}_a\left(c_1(n-1) - c_2\omega^{(b-a)(p-1)} - c_3\omega^{(c-a)(p-1)}\right) + \\ &\tilde{\mathbf{h}}_b\left(c_2(n-1) - c_1\omega^{(a-b)(p-1)} - c_3\omega^{(c-b)(p-1)}\right) + \\ &\tilde{\mathbf{h}}_c\left(c_3(n-1) - c_1\omega^{(a-c)(p-1)} - c_2\omega^{(b-c)(p-1)}\right) = \lambda c_1\tilde{\mathbf{h}}_a + \lambda c_2\tilde{\mathbf{h}}_b + \lambda c_3\tilde{\mathbf{h}}_c.\end{aligned}$$

If we equate corresponding parts of the l.h.s. and r.h.s. we get the three following equations:

$$\begin{aligned}\lambda c_1 &= c_1(n-1) - c_2\omega^{(b-a)(p-1)} - c_3\omega^{(c-a)(p-1)} \\ \lambda c_2 &= c_2(n-1) - c_1\omega^{(a-b)(p-1)} - c_3\omega^{(c-b)(p-1)} \\ \lambda c_3 &= c_3(n-1) - c_1\omega^{(a-c)(p-1)} - c_2\omega^{(b-c)(p-1)}.\end{aligned}$$

With careful manipulation we get the following:

$$c_1^2\omega^{(a-b)(p-1)} + c_1c_3\omega^{(c-b)(p-1)} = c_2^2\omega^{(b-a)(p-1)} + c_3c_2\omega^{(c-a)(p-1)}$$
$$c_2^2\omega^{(b-c)(p-1)} + c_2c_1\omega^{(a-c)(p-1)} = c_3^2\omega^{(c-b)(p-1)} + c_1c_3\omega^{(a-b)(p-1)}$$
$$c_3^2\omega^{(c-a)(p-1)} + c_3c_2\omega^{(b-a)(p-1)} = c_1^2\omega^{(a-c)(p-1)} + c_2c_1\omega^{(b-c)(p-1)}.$$

We can isolate a variable for each equation to get

$$c_3 = \frac{c_2^2\omega^{(b-a)(p-1)} - c_1^2\omega^{(a-b)(p-1)}}{c_2\omega^{(c-a)(p-1)} - c_1\omega^{(c-b)(p-1)}}$$
$$c_1 = \frac{c_3^2\omega^{(c-b)(p-1)} - c_2^2\omega^{(b-c)(p-1)}}{c_3\omega^{(a-b)(p-1)} - c_2\omega^{(a-c)(p-1)}}$$
$$c_2 = \frac{c_1^2\omega^{(a-c)(p-1)} - c_3^2\omega^{(c-a)(p-1)}}{c_1\omega^{(b-c)(p-1)} - c_3\omega^{(b-a)(p-1)}}.$$

If we take c_3 and substitute into the first eigenvalue equation we get

$$\lambda c_1 = nc_1;$$

therefore, one possible solution is $\lambda = n$. One open question is to show that the largest eigenvalue is n.

4. Steiner Systems

Fickus's construction of ETFs involve special cases of block designs called $(2, k, v)$-Steiner systems. To begin with Fickus states the following facts as a result:

- $\mathbf{A}^T$ is a $\{0, 1\}$-incidence matrix that corresponds to $(2, k, v)$-Steiner system.
- The matrix $\mathbf{A}^T$ is $\frac{v(v-1)}{k(k-1)} \times v$ in dimension.
- Each row has k ones.
- Each column has $\frac{(v-1)}{(k-1)}$ ones.
- The *dot* product of any two columns is exactly 1.

As an example suppose we let $k = 2$ and $v = 4$. Then $\mathbf{A}^T =$

$$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}.$$

Then we construct a $1 + \frac{v-1}{k-1} \times 1 + \frac{v-1}{k-1}$ complex Hadamard matrix.

$$\mathbf{H} = \left[\omega^{(j-1)(k-1)}\right]_{j,k=1}^{1+\frac{v-1}{k-1}}.$$

Then we replace the "1" elements in $\mathbf{A}^T$ with $\frac{v-1}{k-1}$ distinct rows from $\mathbf{H}$. Since there are v columns in $\mathbf{A}^T$ and each "1" is replaced with a row of length $1 + \frac{v-1}{k-1}$, then this will yield a matrix $\mathbf{F}$ that is $\frac{v(v-1)}{k(k-1)} \times v\left(1 + \frac{v-1}{k-1}\right)$.

$$\mathbf{F} = [f_j]_{j=1}^{v\left(1+\frac{v-1}{k-1}\right)}$$

Keep in mind that the elements in $\mathbf{H}$ are unimodular. Since each f_j contains $\frac{v-1}{k-1}$ unimodular entries from $\mathbf{H}$, then

$$\|f_j\| = \sqrt{\frac{v-1}{k-1}}.$$

We shall examine the easy case and "clip" the first row of the matrix $\mathbf{H}$ as a result our inner product space becomes

$$\langle f_j, f_l \rangle = \begin{cases} \frac{v-1}{k-1} & j = l, \\ -1 & j \neq l. \end{cases}$$

Let us also consider the matrix $\mathbf{F}$ in terms of its rows

$$\mathbf{F} = \begin{bmatrix} r_1 \\ \vdots \\ r_{\frac{v(v-1)}{k(k-1)}} \end{bmatrix}$$

and each $r_j \in \mathbb{C}^{v\left(1+\frac{v-1}{k-1}\right)}$. Since each row of $\mathbf{A}^T$ contains k 1's and each 1 is replaced with a row of $\mathbf{H}$ then each row of $\mathbf{F}$ contains $k\left(1+\frac{v-1}{k-1}\right)$ unimodular entries. Notice that. The rows of $\mathbf{F}$ must be orthogonal because when taking the inner product you will take the inner product of rows of 0's or two rows from $\mathbf{H}$ which are necessarily orthogonal. Hence

$$\langle r_j, r_l \rangle = \begin{cases} k\left(1+\frac{v-1}{k-1}\right) & j = l, \\ 0 & j \neq l. \end{cases}$$

Therefore, we have

$$\sum_{j=1}^{v\left(1+\frac{v-1}{k-1}\right)} f_j \otimes f_j = \mathbf{F}\mathbf{F}^* = \Big[\langle r_j, r_l \rangle\Big]_{j,l=1}^{\frac{v(v-1)}{k(k-1)}} = k\left(1+\frac{v-1}{k-1}\right)\mathbf{I}_{\frac{v(v-1)}{k(k-1)}}.$$

This obviously does not meet the hypothesis of MSS. But we can easily rectify this by renormalizing our frame vectors such that

$$\langle f_j, f_l \rangle = \frac{1}{k\left(1+\frac{v-1}{k-1}\right)} \begin{cases} \frac{v-1}{k-1} & j = l, \\ -1 & j \neq l. \end{cases}$$

Or simply

$$\langle f_j, f_l \rangle = \begin{cases} \frac{v-1}{k(k+v-2)} & j = l, \\ \frac{1-k}{k(k+v-2)} & j \neq l. \end{cases}$$

I claim this is true for inner products of vectors in the same *block*. If we have vectors in different blocks we get

$$\langle f_j, f_l \rangle = \begin{cases} \frac{v-1}{k(k+v-2)} & j = l, \\ \pm i\frac{1-k}{k(k+v-2)} \text{ or } \pm\frac{1-k}{k(k+v-2)} & j \neq l. \end{cases}$$

The number of blocks is v, the number of columns in $\mathbf{A}^T$. Perhaps we can have our algorithm form subsets that are in the same blocks?

5. Subsets in the Same Blocks

Let us explore subsets with all vectors in the same block of our $\mathbf{F}$ matrix. As usual let us begin witht the trivial problem of a one element subset and find $\|f_j f_j^*\|$. This is the same as solving

$$f_j f_j^* v = \lambda v.$$

Let $v = c_1 f_j$; therefore,

$$f_j f_j^* c_1 f_j = \lambda c_1 f_j.$$

This becomes

$$c_1 \frac{v-1}{k(k+v-2)} f_j = \lambda c_1 f_j.$$

It is clear that the only two solutions are $\lambda = \{\frac{v-1}{k(k+v-2)}, 0\}$.
Suppose we consider a two element subset, with out loss of generality let $\mathcal{S} = \{1, 2\}$. We need to find $\|f_1 f_1^* + f_2 f_2^*\|$. Thus we solve a similar eigenvalue problem

$$\left(f_1 f_1^* + f_2 f_2^*\right) v = \lambda v.$$

We make a similar substitution $v = c_1 f_1 + c_2 f_2$ to get

$$\left(f_1 f_1^* + f_2 f_2^*\right)\left(c_1 f_1 + c_2 f_2\right) = \lambda\left(c_1 f_1 + c_2 f_2\right)$$

$$c_1 f_1 f_1^* f_1 + c_1 f_2 f_2^* f_1 + c_2 f_1 f_1^* f_2 + c_2 f_2 f_2^* f_2 = \lambda c_1 f_1 + \lambda c_2 f_2$$

$$\left(\frac{v-1}{k(k+v-2)} c_1 + \frac{1-k}{k(k+v-2)} c_2\right) f_1 + \left(\frac{v-1}{k(k+v-2)} c_2 + \frac{1-k}{k(k+v-2)} c_1\right) f_2 = \lambda c_1 f_1 + \lambda c_2 f_2.$$

If we equate parts we get

$$\lambda c_1 = \frac{v-1}{k(k+v-2)} c_1 + \frac{1-k}{k(k+v-2)} c_2$$

$$\lambda c_2 = \frac{v-1}{k(k+v-2)} c_2 + \frac{1-k}{k(k+v-2)} c_1.$$

Thus,

$$\frac{v-1}{k(k+v-2)} c_1 c_2 + \frac{1-k}{k(k+v-2)} c_2^2 = \frac{v-1}{k(k+v-2)} c_2 c_1 + \frac{1-k}{k(k+v-2)} c_1^2.$$

And

$$c_1^2 = c_2^2 \Rightarrow c_1 = \pm c_2.$$

Thus,

$$\lambda = \frac{v-1}{k(k+v-2)} \pm \frac{1-k}{k(k+v-2)}.$$

Therefore, the three possible eigenvalues are

$$\lambda = \left\{\frac{1}{k}, \frac{v-k}{k(k+v-2)}, 0\right\}.$$

If we consider a subset of size containing three elements $\mathcal{S} = \{1, 2, 3\}$ we will arrive

at the following using similar analysis

$$\lambda c_1 = \frac{v-1}{k(k+v-2)}c_1 + \frac{1-k}{k(k+v-2)}(c_2+c_3)$$

$$\lambda c_2 = \frac{v-1}{k(k+v-2)}c_2 + \frac{1-k}{k(k+v-2)}(c_1+c_3)$$

$$\lambda c_3 = \frac{v-1}{k(k+v-2)}c_3 + \frac{1-k}{k(k+v-2)}(c_1+c_2).$$

If we combine equations we get,

$$\frac{c_2+c_3}{c_1} = \frac{c_1+c_3}{c_2} = \frac{c_1+c_2}{c_3}.$$

And

$$c_2^2 + c_3c_2 = c_1^2 + c_3c_1$$

$$c_3^2 + c_3c_1 = c_2^2 + c_2c_1$$

$$c_1^2 + c_2c_1 = c_3^2 + c_2c_3.$$

One solution is $c_2 = c_3 = c_1$ and

$$\lambda = \frac{v-1+2(1-k)}{k(k+v-2)} = \frac{v-2k+1}{k(k+v-2)}.$$

Another solution is $c_3 = -c_1 - c_2$, which yields

$$\lambda = \frac{v-1}{k(k+v-2)} - \frac{1-k}{k(k+v-2)} = \frac{1}{k}.$$

And of course the trivial solution $\lambda = 0$. Therefore, our spectrum is

$$\left\{\frac{1}{k}, \frac{v-2k+1}{k(k+v-2)}, 0\right\}.$$

If we follow similar analysis for $\mathcal{S} = \{1,2,3,4\}$ we will get a similar extended proportion

$$\frac{c_1+c_2+c_3}{c_4} = \frac{c_1+c_2+c_4}{c_3} = \frac{c_1+c_3+c_4}{c_2} = \frac{c_2+c_3+c_4}{c_1}.$$

If can be shown that the two solutions for the coefficients are

$$c_2 = c_3 = c_4 = c_1,$$

and

$$c_4 = -(c_1+c_2+c_3).$$

Substitution of the solutions will yield

$$\lambda = \left\{\frac{1}{k}, \frac{v-3k+2}{k(k+v-2)}, 0\right\}.$$

I have noticed the following pattern in the above analysis

- $\mathcal{S} = \{1,2\} \Rightarrow \sigma = \left\{\frac{1}{k}, \frac{v-1\cdot k+0}{k(k+v-2)}, 0\right\}$
- $\mathcal{S} = \{1,2,3\} \Rightarrow \sigma = \left\{\frac{1}{k}, \frac{v-2\cdot k+1}{k(k+v-2)}, 0\right\}$
- $\mathcal{S} = \{1,2,3,4\} \Rightarrow \sigma = \left\{\frac{1}{k}, \frac{v-3\cdot k+2}{k(k+v-2)}, 0\right\}$.

This applies if we form subsets in the same *blocks* of $\mathbf{F}$. I claim given subset $\mathcal{S}$ from a block of $\mathbf{F}$ will have the corresponding spectral values of the matrix $\mathbf{F}_{\mathcal{S}}$

$$\sigma = \left\{ \frac{1}{k}, \frac{v-(R-1)k+(R-2)}{k(k+v-2)}, 0 \right\},$$

or simply

$$\sigma = \left\{ \frac{1}{k}, \frac{1}{k} - R\frac{k-1}{k(k+v-2)}, 0 \right\}$$

where $R = |\mathcal{S}|$ is the cardinality of a subset.

6. (2,2,4)-Steiner System

Let us examine a Steiner system where $k = 2$ and $v = 4$. We have the same complex Hadamard matrix as before

$$\mathbf{A}^T = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}.$$

The transpose-incidence matrix has $v = 4$ columns and $\frac{v(v-1)}{k(k-1)} = \frac{4(3)}{2(1)} = 6$ rows. Each column of $\mathbf{A}^T$ has $\frac{v-1}{k-1} = \frac{4-1}{2-1} = 3$ elements of 1's. The corresponding Hadamard Matrix is

$$\mathbf{H} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{bmatrix}.$$

Notice that the dimensions of $\mathbf{H}$ is one more than the number of 1's in a column of $\mathbf{A}^T$, or $1 + \frac{v-1}{k-1} = 1 + 3 = 4$. For simplification of calculations we shall examine the case when we clip the first row of $\mathbf{H}$ to get

$$\tilde{\mathbf{H}} = \begin{bmatrix} 1 & -i & -1 & i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{bmatrix}.$$

We replace the 1 elements in each column of $\mathbf{A}^T$ with a distinct row from $\tilde{\mathbf{H}}$ to get

$$\left[\begin{array}{cccc|cccc|cccc|cccc} 1 & -i & -1 & i & 1 & -i & -1 & i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & -1 & 1 & -1 & 0 & 0 & 0 & 0 & 1 & -i & -1 & i & 0 & 0 & 0 & 0 \\ 1 & i & -1 & -i & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & -i & -1 & i \\ 0 & 0 & 0 & 0 & 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & i & -1 & -i & 0 & 0 & 0 & 0 & 1 & -1 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & i & -1 & -i & 1 & i & -1 & -i \end{array}\right].$$

The columns of this matrix are the frame vectors $\mathbf{F} = [f_j]_{j=1}^{16}$. Notice that there are $v\left(1 + \frac{v-1}{k-1}\right) = 4(4) = 16$ vectors. However, $\mathbf{F}\mathbf{F}^* = 8\mathbf{I}$, thus we need to re-normalize our vectors f_j. To do this we shall multiply our large matrix $\mathbf{F}$ by

$$\sqrt{\frac{k-1}{k(k+v-2)}} = \sqrt{\frac{2-1}{2(2+4-2)}} = \frac{1}{\sqrt{8}}.$$

Thus our re-normalized ETF is

$$\left[\begin{array}{cccc|cccc|cccc|cccc}
\frac{1}{\sqrt{8}} & -\frac{i}{\sqrt{8}} & -\frac{1}{\sqrt{8}} & \frac{i}{\sqrt{8}} & \frac{1}{\sqrt{8}} & -\frac{i}{\sqrt{8}} & -\frac{1}{\sqrt{8}} & \frac{i}{\sqrt{8}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{1}{\sqrt{8}} & -\frac{1}{\sqrt{8}} & \frac{1}{\sqrt{8}} & -\frac{1}{\sqrt{8}} & 0 & 0 & 0 & 0 & \frac{1}{\sqrt{8}} & -\frac{i}{\sqrt{8}} & -\frac{1}{\sqrt{8}} & \frac{i}{\sqrt{8}} & 0 & 0 & 0 & 0 \\
\frac{1}{\sqrt{8}} & \frac{i}{\sqrt{8}} & -\frac{1}{\sqrt{8}} & -\frac{i}{\sqrt{8}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{\sqrt{8}} & -\frac{i}{\sqrt{8}} & -\frac{1}{\sqrt{8}} & \frac{i}{\sqrt{8}} \\
0 & 0 & 0 & 0 & \frac{1}{\sqrt{8}} & -\frac{1}{\sqrt{8}} & \frac{1}{\sqrt{8}} & -\frac{1}{\sqrt{8}} & \frac{1}{\sqrt{8}} & -\frac{1}{\sqrt{8}} & \frac{1}{\sqrt{8}} & -\frac{1}{\sqrt{8}} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{1}{\sqrt{8}} & \frac{i}{\sqrt{8}} & -\frac{1}{\sqrt{8}} & -\frac{i}{\sqrt{8}} & 0 & 0 & 0 & 0 & \frac{1}{\sqrt{8}} & -\frac{1}{\sqrt{8}} & \frac{1}{\sqrt{8}} & -\frac{1}{\sqrt{8}} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{\sqrt{8}} & \frac{i}{\sqrt{8}} & -\frac{1}{\sqrt{8}} & -\frac{i}{\sqrt{8}} & \frac{1}{\sqrt{8}} & \frac{i}{\sqrt{8}} & -\frac{1}{\sqrt{8}} & -\frac{i}{\sqrt{8}}
\end{array}\right]$$

$$= \left[\begin{array}{cccc|cccc|cccc|cccc} f_1 & f_2 & f_3 & f_4 & f_5 & f_6 & f_7 & f_8 & f_9 & f_{10} & f_{11} & f_{12} & f_{13} & f_{14} & f_{15} & f_{16} \end{array}\right]$$

The norm of each column f_j is $\sqrt{3/8}$. Interestingly, the inner product of vectors in the same block is always $-1/8$, but the inner product of two vectors in different blocks are $\pm 1/8$ or $\pm i/8$. Is there a pattern to this? Suppose we assign the blocks names

- $[f_1 \dots f_4] = \mathbf{B}_1$
- $[f_5 \dots f_8] = \mathbf{B}_2$
- $[f_9 \dots f_{12}] = \mathbf{B}_3$
- $[f_{13} \dots f_{16}] = \mathbf{B}_4$

Notice that picking a column from one block and finding the inner product with a column in a different block will result in multiplying only one of the elements in each column at the same positions. This is a result of the multiplication of zeros. For instance, $\langle f_4, f_5 \rangle$ only requires us to find the dot product of the first two elements in both f_4 and f_5. Therefore,

$$\langle f_4, f_5 \rangle = \left(\frac{i}{\sqrt{8}}\right)\left(\frac{1}{\sqrt{8}}\right) = \frac{i}{8}.$$

Let us form a matrix that contains all the inner products for the columns of $\mathbf{B}_1$ and $\mathbf{B}_2$

$$\begin{bmatrix} \langle f_1, f_5 \rangle & \langle f_1, f_6 \rangle & \langle f_1, f_7 \rangle & \langle f_1, f_8 \rangle \\ \langle f_2, f_5 \rangle & \langle f_2, f_6 \rangle & \langle f_2, f_7 \rangle & \langle f_2, f_8 \rangle \\ \langle f_3, f_5 \rangle & \langle f_3, f_6 \rangle & \langle f_3, f_7 \rangle & \langle f_3, f_8 \rangle \\ \langle f_4, f_5 \rangle & \langle f_4, f_6 \rangle & \langle f_4, f_7 \rangle & \langle f_4, f_8 \rangle \end{bmatrix} = \begin{bmatrix} \frac{1}{8} & \frac{i}{8} & -\frac{1}{8} & -\frac{i}{8} \\ -\frac{i}{8} & \frac{1}{8} & \frac{i}{8} & -\frac{1}{8} \\ -\frac{1}{8} & -\frac{i}{8} & \frac{1}{8} & \frac{i}{8} \\ \frac{i}{8} & -\frac{1}{8} & -\frac{i}{8} & \frac{1}{8} \end{bmatrix}$$

$$= \frac{1}{8}\left[i^{k-j}\right]_{j=1,k=5}^{4,8}$$

Let us examine the matrix that contains all the inner products for the columns of $\mathbf{B}_1$ and $\mathbf{B}_3$

$$\begin{bmatrix} \langle f_1, f_9 \rangle & \langle f_1, f_{10} \rangle & \langle f_1, f_{11} \rangle & \langle f_1, f_{12} \rangle \\ \langle f_2, f_9 \rangle & \langle f_2, f_{10} \rangle & \langle f_2, f_{11} \rangle & \langle f_2, f_{12} \rangle \\ \langle f_3, f_9 \rangle & \langle f_3, f_{10} \rangle & \langle f_3, f_{11} \rangle & \langle f_3, f_{12} \rangle \\ \langle f_4, f_9 \rangle & \langle f_4, f_{10} \rangle & \langle f_4, f_{11} \rangle & \langle f_4, f_{12} \rangle \end{bmatrix} = \begin{bmatrix} \frac{1}{8} & \frac{i}{8} & -\frac{1}{8} & -\frac{i}{8} \\ -\frac{1}{8} & -\frac{i}{8} & \frac{1}{8} & \frac{i}{8} \\ \frac{1}{8} & \frac{i}{8} & -\frac{1}{8} & -\frac{i}{8} \\ -\frac{1}{8} & -\frac{i}{8} & \frac{1}{8} & \frac{i}{8} \end{bmatrix}$$

7. Subset Norms and Complex Hadamard Matrices

Let us consider the 5×5 Hadamard matrix

$$\mathbf{H} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & e^{-\frac{2i\pi}{5}} & e^{-\frac{4i\pi}{5}} & e^{\frac{4i\pi}{5}} & e^{\frac{2i\pi}{5}} \\ 1 & e^{-\frac{4i\pi}{5}} & e^{\frac{2i\pi}{5}} & e^{-\frac{2i\pi}{5}} & e^{\frac{4i\pi}{5}} \\ 1 & e^{\frac{4i\pi}{5}} & e^{-\frac{2i\pi}{5}} & e^{\frac{2i\pi}{5}} & e^{-\frac{4i\pi}{5}} \\ 1 & e^{\frac{2i\pi}{5}} & e^{\frac{4i\pi}{5}} & e^{-\frac{4i\pi}{5}} & e^{-\frac{2i\pi}{5}} \end{bmatrix}.$$

Below are the inner products for a clipped Hadamard matrix $\tilde{\mathbf{H}}_q$

$$\langle \tilde{\mathbf{h}}_k, \tilde{\mathbf{h}}_l \rangle = \begin{cases} n-1 & k = l, \\ -\omega^{(k-l)\cdot q} & k \neq l, \end{cases} \tag{7.1}$$

where $q = p - 1$ and p is the p-th row deleted from $\mathbf{H}$. Suppose we wish to find the subset norm corresponding to two vectors $\mathcal{S} = \{a, b\}$ where a, b are the clipped columns $\tilde{\mathbf{h}}_a, \tilde{\mathbf{h}}_b$. In stead of considering the columns as being clipped, I shall consider the columns as having the p element replaced with a 0. For instance we replace the $p = 2$ row with zeros we have

$$\tilde{\mathbf{h}}_3 = \begin{bmatrix} 1 \\ 0 \\ e^{\frac{2i\pi}{5}} \\ e^{-\frac{2i\pi}{5}} \\ e^{\frac{4i\pi}{5}} \end{bmatrix} = \mathbf{h}_3 - \mathbf{H}_{2,3}\mathbf{e}_2 = \mathbf{h}_3 - \mathbf{e}_2^T\mathbf{H}\mathbf{e}_3\mathbf{e}_2.$$

Therefore replacing the p-th row with zeros yields

$$\tilde{\mathbf{h}}_a = \mathbf{h}_a - \mathbf{H}_{p,a}\mathbf{e}_p = \mathbf{h}_a - \mathbf{e}_p^T\mathbf{H}\mathbf{e}_a\mathbf{e}_p.$$

Notice

$$\begin{aligned} \langle \tilde{\mathbf{h}}_a, \tilde{\mathbf{h}}_b \rangle &= \langle \mathbf{h}_a - \mathbf{H}_{p,a}\mathbf{e}_p, \mathbf{h}_b - \mathbf{H}_{p,b}\mathbf{e}_p \rangle \\ &= \langle \mathbf{h}_a, \mathbf{h}_b \rangle + \langle \mathbf{h}_a, -\mathbf{H}_{p,b}\mathbf{e}_p \rangle + \langle -\mathbf{H}_{p,a}\mathbf{e}_p, \mathbf{h}_b \rangle + \langle -\mathbf{H}_{p,a}\mathbf{e}_p, -\mathbf{H}_{p,b}\mathbf{e}_p \rangle. \end{aligned}$$

Notice that the first inner product must be 0 because the columns of $\mathbf{H}$ are orthogonal; hence,

$$\begin{aligned} &= \langle \mathbf{h}_a, -\mathbf{H}_{p,b}\mathbf{e}_p \rangle + \langle -\mathbf{H}_{p,a}\mathbf{e}_p, \mathbf{h}_b \rangle + \langle -\mathbf{H}_{p,a}\mathbf{e}_p, -\mathbf{H}_{p,b}\mathbf{e}_p \rangle \\ &= \overline{\mathbf{H}}_{p,b}\langle \mathbf{h}_a, -\mathbf{e}_p \rangle + \mathbf{H}_{p,a}\langle -\mathbf{e}_p, \mathbf{h}_b \rangle + \mathbf{H}_{p,a}\overline{\mathbf{H}}_{p,b} \\ &= -\overline{\mathbf{H}}_{p,b}\mathbf{H}_{p,a} - \mathbf{H}_{p,a}\overline{\mathbf{H}}_{p,b} + \mathbf{H}_{p,a}\overline{\mathbf{H}}_{p,b} \\ &= -\overline{\mathbf{H}}_{p,b}\mathbf{H}_{p,a} \\ &= -\left(\overline{\omega}^{(p-1)(b-1)}\right)\left(\omega^{(p-1)(a-1)}\right) \\ &= -\left(\omega^{-(p-1)(b-1)}\right)\left(\omega^{(p-1)(a-1)}\right) \\ &= -\omega^{((a-1)-(b-1))(p-1)} = -\omega^{(a-b)(p-1)}. \end{aligned}$$

This yields the same result as we for the *clipped* columns. Therefore, I claim that the inner product space for a set of clipped column vectors is the same as the inner product space of a set of column vector with the same entry position replaced with 0. Therefore, if we replace the p-th row of $\mathbf{H}$ with zeros, then

$$\tilde{\mathbf{h}}_j = \left[\mathbf{H}_{k,j}\delta_{kp}\right]_{k=1}^{n},$$

where δ_{kp} is the Kronecker delta

$$\delta_{kp} = \begin{cases} 0 & k = p, \\ 1 & k \neq p. \end{cases}$$

The inner product remains

$$\langle \tilde{\mathbf{h}}_k, \tilde{\mathbf{h}}_l \rangle = \begin{cases} n-1 & k = l, \\ -\omega^{(k-l)\cdot q} & k \neq l, \end{cases} \tag{7.2}$$

where $q = p - 1$.
Suppose we find the subset norm corresponding to a two element subset and with out loss of generality let $\mathcal{S} = 1, 2$. This means calculating

$$\|\tilde{\mathbf{h}}_1 \otimes \tilde{\mathbf{h}}_1 + \tilde{\mathbf{h}}_2 \otimes \tilde{\mathbf{h}}_2\| = ?$$

Keep in mind this time I am considering the columns $\tilde{\mathbf{h}}_j$ not clipped, but having one element replaced with a zero, for the p-th row. This requires solving the eigenvalue problem

$$\left(\tilde{\mathbf{h}}_1\tilde{\mathbf{h}}_1^* + \tilde{\mathbf{h}}_2\tilde{\mathbf{h}}_2^*\right) v = \lambda v.$$

Let $v = c_1\tilde{\mathbf{h}}_1 + c_2\tilde{\mathbf{h}}_2$, and we have

$$\left(\tilde{\mathbf{h}}_1\tilde{\mathbf{h}}_1^* + \tilde{\mathbf{h}}_2\tilde{\mathbf{h}}_2^*\right)\left(c_1\tilde{\mathbf{h}}_1 + c_2\tilde{\mathbf{h}}_2\right) = \lambda\left(c_1\tilde{\mathbf{h}}_1 + c_2\tilde{\mathbf{h}}_2\right).$$

This becomes

$$c_1\tilde{\mathbf{h}}_1\tilde{\mathbf{h}}_1^*\tilde{\mathbf{h}}_1 + c_1\tilde{\mathbf{h}}_2\tilde{\mathbf{h}}_2^*\tilde{\mathbf{h}}_1 + c_2\tilde{\mathbf{h}}_1\tilde{\mathbf{h}}_1^*\tilde{\mathbf{h}}_2 + c_2\tilde{\mathbf{h}}_2\tilde{\mathbf{h}}_2^*\tilde{\mathbf{h}}_2 = \lambda c_1\tilde{\mathbf{h}}_1 + \lambda c_2\tilde{\mathbf{h}}_2.$$

Substitute the inner product values to get

$$c_1(n-1)\tilde{\mathbf{h}}_1 - c_1\tilde{\mathbf{h}}_2\omega^{(1-2)q} - c_2\tilde{\mathbf{h}}_1\omega^{(2-1)q} + c_2(n-1)\tilde{\mathbf{h}}_2 = \lambda c_1\tilde{\mathbf{h}}_1 + \lambda c_2\tilde{\mathbf{h}}_2.$$

With rearrangement we get

$$\tilde{\mathbf{h}}_1\left(c_1(n-1) - c_2\omega^{(2-1)q}\right) + \tilde{\mathbf{h}}_2\left(c_2(n-1) - c_1\omega^{(1-2)q}\right) = \lambda c_1\tilde{\mathbf{h}}_1 + \lambda c_2\tilde{\mathbf{h}}_2.$$

We equate corresponding parts we get

$$\lambda c_1 = c_1(n-1) - c_2\omega^{(2-1)q}$$
$$\lambda c_2 = c_2(n-1) - c_1\omega^{(1-2)q}.$$

Then

$$(n-1) - \frac{c_2}{c_1}\omega^{(2-1)q} = (n-1) - \frac{c_1}{c_2}\omega^{(1-2)q},$$

and

$$c_2^2\omega^{(2-1)q} = c_1^2\omega^{(1-2)q}.$$

Which is the same as

$$c_2^2\omega^{(2-1)q} = c_1^2\omega^{-(2-1)q}.$$

Notice if we let $p = 1$ (this corresponds to replacing the first row with zeros) we get the same result as shown earlier. Here it is not as obvious. We get

$$c_2^2 = c_1^2 \omega^{-2(2-1)q},$$

which yields

$$c_2 = \pm c_1 \omega^{(1-2)q}.$$

If we make the substitution into

$$\lambda = (n-1) - \frac{c_2}{c_1}\omega^{(2-1)q},$$

we get

$$\lambda = (n-1) \pm \omega^{(2-1)q}\omega^{(1-2)q}.$$

Which gives the same result as shown earlier

$$\lambda = n - 1 \pm 1$$

The corresponding eigenvectors are

$$v = \left\{ c_1 \tilde{\mathbf{h}}_1 \pm c_1 \omega^{(1-2)q} \tilde{\mathbf{h}}_2 \right\}$$

I claim given a subset of two vectors $\{\tilde{\mathbf{h}}_a, \tilde{\mathbf{h}}_b\}$ the solution to the eigenvalue problem

$$\left(\tilde{\mathbf{h}}_a \tilde{\mathbf{h}}_a^* + \tilde{\mathbf{h}}_b \tilde{\mathbf{h}}_b^* \right) v = \lambda v$$

has the corresponding eigenvalues and vectors are

$$\sigma = \{n, n-2, 0\}$$

and

$$v = \left\{ \tilde{\mathbf{h}}_a \pm \omega^{(a-b)q} \tilde{\mathbf{h}}_b, \mathbf{0} \right\}$$

Suppose we now examine the case where there are three elements in a subset $\mathcal{S} = \{1, 2, 3\}$.

$$\left(\tilde{\mathbf{h}}_1 \tilde{\mathbf{h}}_1^* + \tilde{\mathbf{h}}_2 \tilde{\mathbf{h}}_2^* + \tilde{\mathbf{h}}_3 \tilde{\mathbf{h}}_3^* \right) \left(c_1 \tilde{\mathbf{h}}_1 + c_2 \tilde{\mathbf{h}}_2 + c_3 \tilde{\mathbf{h}}_3 \right) = \lambda \left(c_1 \tilde{\mathbf{h}}_1 + c_2 \tilde{\mathbf{h}}_2 + c_3 \tilde{\mathbf{h}}_3 \right).$$

The left hand side becomes

$$\begin{aligned}
&c_1 \tilde{\mathbf{h}}_1 \tilde{\mathbf{h}}_1^* \tilde{\mathbf{h}}_1 + c_1 \tilde{\mathbf{h}}_2 \tilde{\mathbf{h}}_2^* \tilde{\mathbf{h}}_1 + c_1 \tilde{\mathbf{h}}_3 \tilde{\mathbf{h}}_3^* \tilde{\mathbf{h}}_1 + \\
&c_2 \tilde{\mathbf{h}}_1 \tilde{\mathbf{h}}_1^* \tilde{\mathbf{h}}_2 + c_2 \tilde{\mathbf{h}}_2 \tilde{\mathbf{h}}_2^* \tilde{\mathbf{h}}_2 + c_2 \tilde{\mathbf{h}}_3 \tilde{\mathbf{h}}_3^* \tilde{\mathbf{h}}_2 + \\
&c_3 \tilde{\mathbf{h}}_1 \tilde{\mathbf{h}}_1^* \tilde{\mathbf{h}}_3 + c_3 \tilde{\mathbf{h}}_2 \tilde{\mathbf{h}}_2^* \tilde{\mathbf{h}}_3 + c_3 \tilde{\mathbf{h}}_3 \tilde{\mathbf{h}}_3^* \tilde{\mathbf{h}}_3 \\
&= \\
&\tilde{\mathbf{h}}_1 \left(c_1(n-1) - c_2 \omega^{(2-1)q} - c_3 \omega^{(3-1)q} \right) + \\
&\tilde{\mathbf{h}}_2 \left(-c_1 \omega^{(1-2)q} + c_2(n-1) - c_3 \omega^{(3-2)q} \right) + \\
&\tilde{\mathbf{h}}_3 \left(-c_1 \omega^{(1-3)q} - c_2 \omega^{(2-3)q} + c_3(n-1) \right).
\end{aligned}$$

If we equate parts with the right hand side we get

$$\begin{aligned}
\lambda c_1 &= c_1(n-1) - c_2 \omega^{(2-1)q} - c_3 \omega^{(3-1)q} \\
\lambda c_2 &= -c_1 \omega^{(1-2)q} + c_2(n-1) - c_3 \omega^{(3-2)q} \\
\lambda c_3 &= -c_1 \omega^{(1-3)q} - c_2 \omega^{(2-3)q} + c_3(n-1).
\end{aligned}$$

We can obtain the following extended proportion

$$\frac{c_2\omega^{(2-1)q}+c_3\omega^{(3-1)q}}{c_1}=\frac{c_1\omega^{(1-2)q}+c_3\omega^{(3-2)q}}{c_2}=\frac{c_1\omega^{(1-3)q}+c_2\omega^{(2-3)q}}{c_3}.$$

One solution set is

$$c_1=\left(\omega^{(2-1)+2(3-1)+(3-2)}\right)^{q/3}c_3=\omega^{2q}c_3,$$

$$c_2=\left(\omega^{(1-2)+(3-1)+2(3-2)}\right)^{q/3}c_3=\omega^{q}c_3.$$

Another is

$$c_1=(-1)^{2/3}\left(\omega^{(2-1)+2(3-1)+(3-2)}\right)^{q/3}c_3=(-1)^{2/3}\omega^{2q}c_3,$$

$$c_2=-(-1)^{1/3}\left(\omega^{(1-2)+(3-1)+2(3-2)}\right)^{q/3}c_3=-(-1)^{1/3}\omega^{q}c_3.$$

Another is

$$c_1=-(-1)^{1/3}\left(\omega^{(2-1)+2(3-1)+(3-2)}\right)^{q/3}c_3=-(-1)^{1/3}\omega^{2q}c_3,$$

$$c_2=(-1)^{2/3}\left(\omega^{(1-2)+(3-1)+2(3-2)}\right)^{q/3}c_3=(-1)^{2/3}\omega^{q}c_3.$$

With substitution of the first solution set into

$$\lambda=-\frac{c_1}{c_3}\omega^{(1-3)q}-\frac{c_2}{c_3}\omega^{(2-3)q}+(n-1),$$

we get

$$\lambda=-1-1+n-1=n-3.$$

If we substitute the other two solutions we get

$$\begin{aligned}\lambda&=-(-1)^{2/3}+(-1)^{1/3}+n-1\\&=-e^{\frac{2\pi i}{3}}+e^{\frac{\pi i}{3}}+n-1\\&=-\left(\cos\frac{2\pi}{3}+i\sin\frac{2\pi}{3}\right)+\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}+n-1\\&=-\left(-\frac{1}{2}+\frac{i\sqrt{3}}{2}\right)+\frac{1}{2}+\frac{i\sqrt{3}}{2}+n-1\\&=\frac{1}{2}-\frac{i\sqrt{3}}{2}+\frac{1}{2}+\frac{i\sqrt{3}}{2}+n-1\\&=n.\end{aligned}$$

Therefore, the spectral values are $n, n-3, 0$. This analysis was convenient because we examined the subset $\mathcal{S}=\{1,2,3\}$. Let us consider the generic case $\mathcal{S}=\{\alpha,\beta,\gamma\}$. This will yield the following solutions

$$c_\alpha=\left(\omega^{(\beta-\alpha)+2(\gamma-\alpha)+(\gamma-\beta)}\right)^{q/3}c_\gamma=\omega^{(\gamma-\alpha)q}c_\gamma,$$

$$c_\beta=\left(\omega^{(\alpha-\beta)+(\gamma-\alpha)+2(\gamma-\beta)}\right)^{q/3}c_\gamma=\omega^{(\gamma-\beta)q}c_\gamma.$$

Similarly,

$$c_\alpha = (-1)^{2/3}\,\omega^{(\gamma-\alpha)q}c_\gamma,$$

$$c_\beta = -(-1)^{1/3}\,\omega^{(\gamma-\beta)q}c_\gamma;$$

and,

$$c_\alpha = -(-1)^{1/3}\,\omega^{(\gamma-\alpha)q}c_\gamma,$$

$$c_\beta = (-1)^{2/3}\,\omega^{(\gamma-\beta)q}c_\gamma.$$

With substitution of the first solution set into

$$\lambda = -\frac{c_\alpha}{c_\gamma}\omega^{(\alpha-\gamma)q} - \frac{c_\beta}{c_\gamma}\omega^{(\beta-\gamma)q} + (n-1),$$

we get again $\lambda = n-3$. If we substitute either of the other two solutions we get

$$\begin{aligned}\lambda &= -(-1)^{2/3} + (-1)^{1/3} + n - 1\\ &= n\end{aligned}$$